Exp 7: Anomaly Detection: Implement an anomaly detection system for detecting outliers in data (e.g., fraud detection) and perform comparative analysis. (Isolation Forest, Local Outlier Factor, One-Class SVM)

Note: Use Evaluation Metrics (Accuracy, Precision, Recall, F1-Score for binary classification), (ROC-AUC, Accuracy, Precision, Recall, Weighted / Macro/ Micro Average F1-Score for Multiclassification task) and Confusion Matrix.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.ensemble import IsolationForest
from sklearn.neighbors import LocalOutlierFactor
from sklearn.svm import OneClassSVM
from sklearn.metrics import accuracy_score,precision_score,recall_score,f1_score,confusion_matrix,roc_au
import pickle
```

```
df=pd.read_csv('/content/drive/MyDrive/anomaly_detection_dataset_500_samples.csv')
df.head()
```

	Feature_1	Feature_2	Anomaly_Label
0	43.259073	53.716320	0
1	43.815246	43.397717	0
2	46.942411	42.966695	0
3	48.911594	55.493884	0
4	52.575238	69.263657	0

Next steps: Generate code with df New interactive sheet

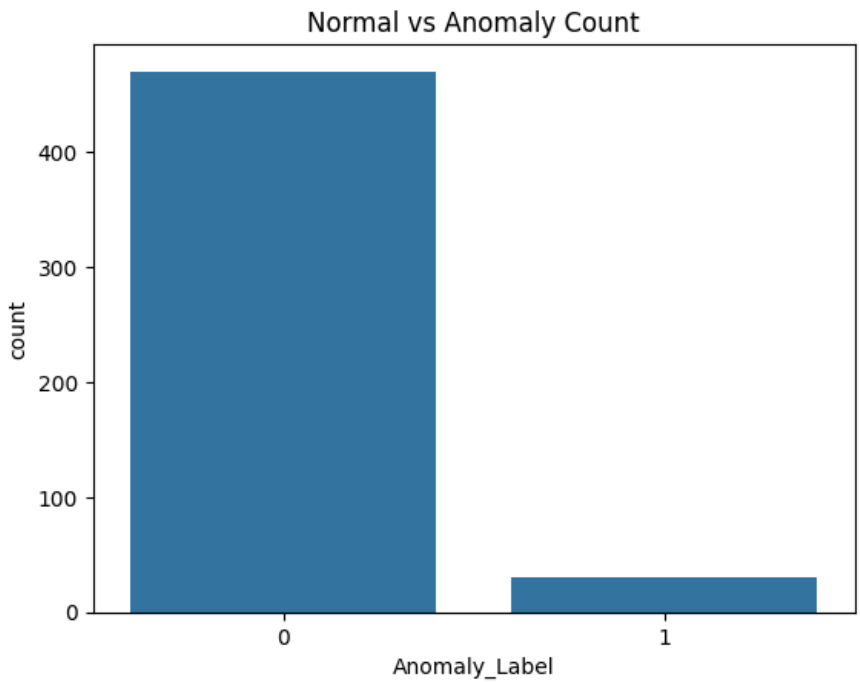
```
print(df.shape)
df.info()
df.describe()
```

```
(500, 3)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 3 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Feature_1      500 non-null    float64
 1   Feature_2      500 non-null    float64
 2   Anomaly_Label  500 non-null    int64
dtypes: float64(2), int64(1)
memory usage: 11.8 KB
```

	Feature_1	Feature_2	Anomaly_Label
count	500.000000	500.000000	500.000000
mean	52.422862	52.573850	0.060000
std	10.600455	10.641964	0.237725
min	33.793663	38.490394	0.000000
25%	47.027724	47.097147	0.000000
50%	50.447020	50.566493	0.000000
75%	53.827497	54.163940	0.000000
max	95.539911	95.402821	1.000000

```
sns.countplot(x="Anomaly_Label",data=df)
plt.title("Normal vs Anomaly Count")
```

```
plt.show()
```




```
X=df[["Feature_1","Feature_2"]]
y=df["Anomaly_Label"]
```

```
sc=StandardScaler()
X_scaled=sc.fit_transform(X)
```

```
X_train,X_test,y_train,y_test=train_test_split(X_scaled,y,test_size=0.2,random_state=0)
```

```
models={
"Isolation Forest":IsolationForest(contamination=0.1,random_state=0),
"Local Outlier Factor":LocalOutlierFactor(novelty=True),
"One-Class SVM":OneClassSVM(nu=0.1)
}
```

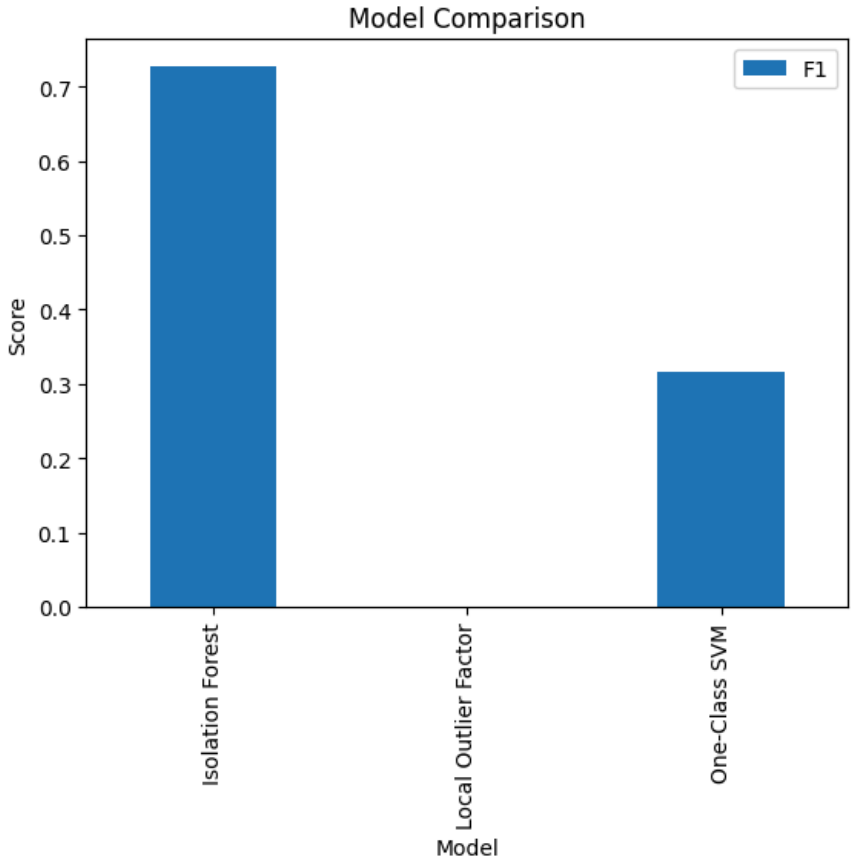
```
results=[]
for name,model in models.items():
    model.fit(X_train)
    pred=model.predict(X_test)
    pred=np.where(pred==-1,1,0)
    acc=accuracy_score(y_test,pred)
    pre=precision_score(y_test,pred)
    rec=recall_score(y_test,pred)
    f1=f1_score(y_test,pred)
    auc=roc_auc_score(y_test,pred)
    results.append([name,acc,pre,rec,f1,auc])
results_df=pd.DataFrame(results,columns=["Model","Accuracy","Precision","Recall","F1","ROC_AUC"])
results_df
```

	Model	Accuracy	Precision	Recall	F1	ROC_AUC
0	Isolation Forest	0.94	0.571429	1.000	0.727273	0.967391
1	Local Outlier Factor	0.88	0.000000	0.000	0.000000	0.478261
2	One-Class SVM	0.87	0.272727	0.375	0.315789	0.644022

Next steps: Generate code with results_df New interactive sheet

```
results_df.plot(x="Model",y="F1",kind="bar")
plt.title("Model Comparison")
```

```
plt.ylabel("Score")
plt.show()
```

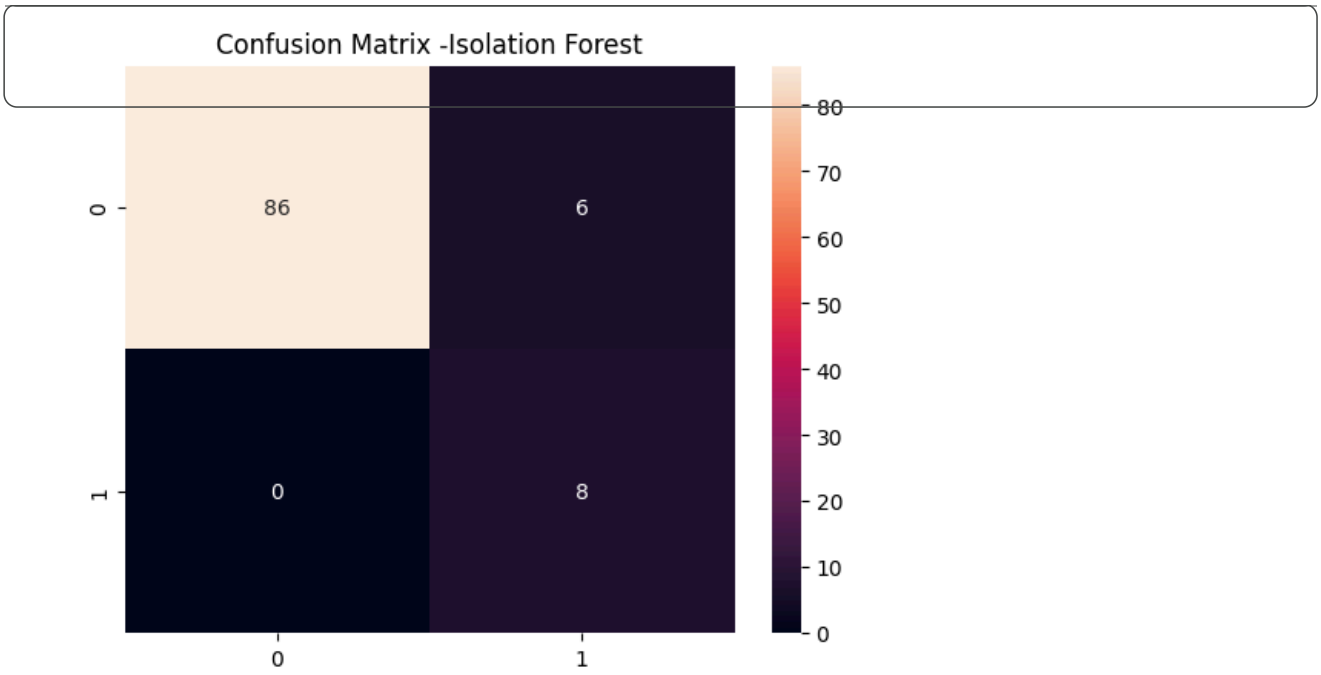



```
best=results_df.sort_values("F1",ascending=False).iloc[0,0]
best_model=models[best]
print("Best Model:",best)
```

```
Best Model: Isolation Forest
```

```
with(open("best_model.pkl","wb")) as f:
    pickle.dump(best_model,f)
```

```
pred=best_model.predict(X_test)
pred=np.where(pred==-1,1,0)
cm=confusion_matrix(y_test,pred)
sns.heatmap(cm,annot=True,fmt='d')
plt.title("Confusion Matrix -"+best)
plt.show()
```


Confusion Matrix -Isolation Forest
86
6
0
8
0
1
0
1
80
70
60
50
40
30
20
10
0